

Multimodal Cross-Document Event Coreference Resolution Using Linear Semantic Transfer and Mixed-Modality Ensembles

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Abstract

Event coreference resolution (ECR) is the task of determining when distinct mentions of events within a multi-document corpus are actually linked to the same underlying occurrence. Images of the events can help facilitate resolution when language is ambiguous. Here, we propose a multimodal cross-document event coreference resolution method that integrates visual and textual cues with a simple linear map between vision and language models. As existing ECR benchmark datasets rarely provide images for all event mentions, we augment the popular ECB+ dataset with event-centric images scraped from the internet and generated using image diffusion models. We establish three methods that incorporate images and text for coreference: 1) a standard fused model with finetuning, 2) a novel linear mapping method without finetuning and 3) an ensembling approach based on splitting mention pairs by semantic and discourse-level difficulty. We evaluate on 2 datasets: the augmented ECB+, and AIDA Phase 1. Our ensemble systems using cross-modal linear mapping establish an upper limit (91.9 CoNLL F1) on ECB+ ECR performance given the preprocessing assumptions used, and establish a novel baseline on AIDA Phase 1. Our results demonstrate the utility of multimodal information in ECR for certain challenging coreference problems, and highlight a need for more multimodal resources in the coreference resolution space.

Keywords: event coreference resolution, multimodality, ensemble methods

1. Introduction

Imagine two newspaper articles about the same event. The articles come from different sources with radically different perspectives and report the event with very different language. They use different action verbs, include ambiguous pronominal references, describe causes differently, and even attribute different intentionality to the event—for example, “*Buzina, 45, was shot dead*” vs. “*He was murdered*”. An automated system may be unable to identify from the text alone that the two events described are actually the same. This is the problem of *cross-document coreference resolution* (CDCR) of events: inferring that two event mentions in different documents actually refer to the same thing.

Now imagine that each of the articles is accompanied by an image. While not identical, they clearly contain the same people, entities, and actions. This would be strong evidence to a reader that the two events described in the different articles are in fact the same.

Purely text-based approaches to CDCR, while built on sophisticated Transformer-based language models (LMs) (Vaswani et al., 2017; Beltagy et al., 2020), are blind to such potentially useful multimodal information. This problem is exacerbated by the relative dearth of multimodal information included in event CDCR corpora.

In this work, we propose a novel multimodal event CDCR method. Where current state-of-the-art coreference approaches that consider visual information demonstrate the utility of a multimodal approach, they do so at a high computational cost (Guo et al., 2022). Furthermore, they typi-

cally focus on linking objects rather than events. We address the sparsity of multimodal data in benchmark datasets by retrieving images associated with the metadata of event mentions, and generating event-centric images with state-of-the-art image diffusion models. We perform coreference experiments in a fully multimodal setting and rigorously test the contribution of multimodal information to CDCR.

In total, our novel contributions include:

- A novel approach to multimodal *cross document event coreference* (MM-CDCR) including a low-compute, bidirectional *linear semantic transfer* technique (Lin-Sem) based on *semantic equivalence* across modalities;
- A model ensemble hybrid approach that applies text-only or multimodal methods to different categories of mention pairs based on their semantic and discourse-level difficulty;
- A novel method for enriching text-only coreference datasets (e.g., ECB+ (Cybulska and Vossen, 2014)) with event-centric images using generative image diffusion;
- A new benchmark result on the AIDA Phase 1 dataset (Tracey et al., 2022), an explicitly multimodal event CDCR dataset. To our knowledge, this is the first evaluation performed over this dataset.

2. Related Work

Cross-Document Event Coreference Resolution Most previous works on CDCR have been limited to *text-only* Eisenstein and Davis (2006); Chen et al. (2011). Early works (e.g., Humphreys et al. (1997); Bagga and Baldwin (1999); Chen

and Ji (2009)) used supervised training over features like part-of-speech tags, phrasal-matching, or aligned arguments. While Kenyon-Dean et al. (2018) enhanced lexical features with “static” embeddings like contextual word2vec (Mikolov et al., 2013), most recent works (Yu et al., 2022; Caciularu et al., 2021; Yadav et al., 2021) uses latent representations from Transformer-based encoders to compute pairwise mention scores of possible antecedents. Works such as Held et al. (2021) and Ahmed et al. (2023) overcome the quadratic complexity of the mention pair architecture by pruning negative pairs using discourse-coherence and lexical similarity (synonymous lemma pairs) respectively. We use Ahmed et al. (2023)’s “oracle” assumption for our pruning procedure.

Multimodal Frameworks Most previous works in multimodal vision-language processing (e.g., (Le et al., 2019; Tan and Bansal, 2019)) have been compute-intensive, using separate encoders for visual and linguistic inputs, and auxiliary encoders for cross-modal or query-related modeling. High-performing but high-compute models like ViL-BERT (Lu et al., 2019) concatenate embeddings from different modalities before fine-tuning. Works such as Li et al. (2020), Tong et al. (2020), and Chen et al. (2021) leverage a common representation space for coreference-adjacent tasks like event extraction and detection in images and videos, but emphasize finding relations within a document or a topic. Works specific to multi-modal *entity* coreference resolution such as Guo et al. (2022) treat it largely as a grounding problem, using graph networks to link references in dialogue to items in a scene before feeding representations into BERT-style encoders to resolve scene-based visual-linguistic coreference chains. Our work is multimodal, cross-document, and event focused, and performs faster with the aid of linear mappings.

Linear Projection Across Neural Networks

Previous research within computer vision has explored using affine (McNeely-White et al., 2020, 2022) as well as non-linear (Lenc and Vedaldi, 2015) transformations to explore equivalence of *unimodal* function approximators like CNNs. They show that two distinct, highly non-linear neural networks can learn similar properties transferable up to a linear projection while retaining near-equivalent performance on tasks like image classification or facial recognition. Similar techniques using affine mappings were reported in Merullo et al. (2023), who explore the equivalence of such approximators *across* modalities while also casting new light on high-fidelity transfer of non-linguistic features into a generative LLM via uni-

directional linear projections from image spaces. Nath et al. (2022) demonstrated linear mappings also preserve information across language models. Ghaffari and Krishnaswamy (2023) showed the same between language models and neural networks trained over tabular data.

We use a low-compute, cross-modal, *bidirectional* linear-mapping technique (**Lin-Sem: Linear Semantic Transfer**) between language and vision Transformers, on the challenging event coreference task. We demonstrate where this linear transfer is providing useful information toward coreference resolution compared to a text-only discriminative LLM, or fused modality models following standard fine-tuning.

3. Methodology

Fig. 1 illustrates the pipeline for our methodology, the components of which are detailed as follows.

Semantic Equivalence

$$\mathcal{V}(x, y, \phi(x, y)): \mathbb{R}^{n \times w \times h \times 3} \rightarrow \mathbb{R}^{n \times H} \quad (1)$$

$$\mathcal{LLM}(x, y, \phi(x, y)): \mathbb{R}^{n \times m} \rightarrow \mathbb{R}^{n \times H} \quad (2)$$

Let (1) and (2) represent the heterogeneous image and text representations for vision and text Transformer models respectively. $(x, y) \in \chi$ represents all the pairs of samples in sample space χ , $\phi(x, y)$ represents the concatenation of the image or text pair in their respective modalities, n and H represent the total sample pairs and hidden dimensions respectively, and m is the LLM’s max token-length.

We define cross-modal semantic equivalence as follows: two representations $\mathcal{V}$ and $\mathcal{LLM}$ in distinct modalities are semantically equivalent if there exists a bidirectional map $M_{\mathcal{V} \leftrightarrow \mathcal{LLM}}$ s.t.:

$$\forall x, y \in \chi: \mathcal{V}(x, y, \phi(x, y)) \approx M_{\mathcal{LLM} \rightarrow \mathcal{V}} \mathcal{LLM}(x, y, \phi(x, y)) \quad (3)$$

$$\forall x, y \in \chi: \mathcal{LLM}(x, y, \phi(x, y)) \approx M_{\mathcal{V} \rightarrow \mathcal{LLM}} \mathcal{V}(x, y, \phi(x, y)) \quad (4)$$

while assuming both $\mathcal{V}$ and $\mathcal{LLM}$ to be bijective or invertible, so,

$$M_{\mathcal{LLM} \rightarrow \mathcal{V}} = \mathcal{V}(x, y, \phi(x, y)) \circ \mathcal{LLM}(x, y, \phi(x, y))^{-1} \quad (5)$$

$$M_{\mathcal{V} \rightarrow \mathcal{LLM}} = \mathcal{LLM}(x, y, \phi(x, y)) \circ \mathcal{V}(x, y, \phi(x, y))^{-1} \quad (6)$$

Since a closed-form solution to analytically derive the mapping function $M_{\mathcal{V} \leftrightarrow \mathcal{LLM}}$ is not always feasible and since many task-based fine-tuning heads over a Transformer-based LLM involve fitting a linear classification layer, we propose a parameter-efficient linear-mapping technique **Lin-Sem**. We estimate the mapping function within an empirical risk minimization framework by using a ridge regression between the two cross-modal representations. Mathematically,

$$M_{\mathcal{LLM} \rightarrow \mathcal{V}} \leftarrow \text{minimize } ((\mathcal{V} - \beta \mathcal{LLM})^T (\mathcal{V} - \beta \mathcal{LLM}) + \lambda \beta^T \beta) \quad (7)$$

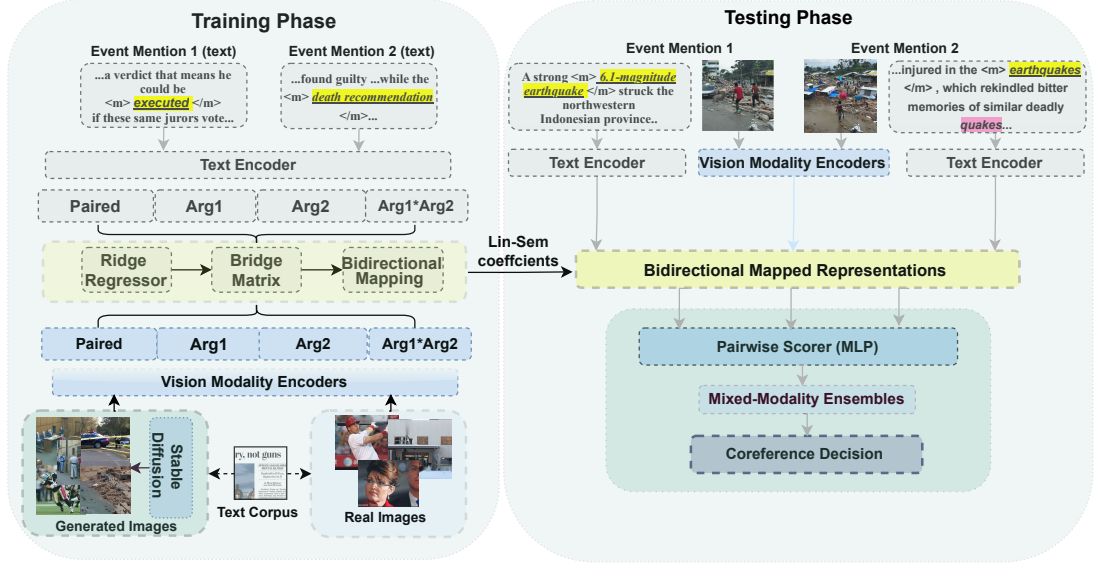


Figure 1: Our approach for Multimodal CDCR using Lin-Sem. Linear Mapping (Lin-Sem) procedure between the distinct text and image embedding spaces for an event pair in the ECB+ corpus. $Arg1$ and $Arg2$ refer to the individual images in the pair and the trigger events (in yellow) surrounded by the $\langle m \rangle$ and $\langle /m \rangle$ special tokens embedded in the text-encoder (LLM).

$$M_{V \rightarrow \mathcal{L}\mathcal{L}\mathcal{M}} \leftarrow \text{minimize } ((\mathcal{L}\mathcal{L}\mathcal{M} - \beta V)^T (\mathcal{L}\mathcal{L}\mathcal{M} - \beta V) + \lambda \beta^T \beta) \quad (8)$$

We assume $\lambda=1$ while β represents the L^2 -norm regularization parameter.

Datasets We evaluated our methods on the ECB+ (Cybulska and Vossen, 2014) and the AIDA Phase 1 (Tracey et al., 2022) datasets. While the former is a popular, English-only CDCR benchmark containing a diverse range of news articles, the latter contains multimodal resources specific to Russia-Ukraine relations, in English, Russian, and Ukrainian. We focus only on the English documents.¹ For our experiments, we used training and evaluation splits following Cybulska and Vossen (2015) for ECB+ and Tracey et al. (2022) for AIDA Phase 1. Table 1 shows corpus-level statistics for these two datasets.

Categorizing Mention Pair Difficulty To empirically evaluate the contribution of cross-modal information toward resolving challenging event mention pairs, we used the gold-standard coreference labels to categorize unseen pairs at inference as

¹The AIDA Phase 1 dataset was created for the DARPA Active Interpretation of Disparate Alternatives (AIDA) program and is available from the Linguistic Data Consortium (catalog number LDC2019E77). It is the only published ECR benchmark that contains multimodal resources specific to cross-document coreference. Events here are specifically in the domain of Russia-Ukraine relations and annotated based on both saliency and the potential for conflicting perspectives.

Split	ECB+			AIDA Phase 1	
	Train	Dev	Test	Practice	Eval
Docs	594	196	206	63	69
Event Mentions	3808	1245	1780	603	846
Clusters	1464	409	805	186	270
Singletons	1053	280	623	132	197
Images	3808*	1245*	1780*	417	662

Table 1: ECB+ and AIDA corpus-level statistics. Tracey et al. (2022) refers to the provided train and test sets as “practice” and “eval”, respectively.

*Including images generated using Stable Diffusion.

easy or *hard* based on semantic and discourse-level similarities. For semantic similarities, we computed Wu-Palmer Similarity (Wu and Palmer, 1994), and the average of the cosine similarities of the two LLM event pair representations encoded in both directions, and added these values. For discourse-level similarities, metadata in both datasets provides information about within-topic and within-document events which we used to score event similarities. For instance, an event pair within the same document and topic would get the highest discourse-level similarity score. These summed scores were then bucketed into *easy* and *hard* semantic transfer categories based on the means of coreferring and non-coreferring samples (see Fig. 2). One “hard” mention pair from ECB+, involving pronominal coreference, is (1) “*In a move $\langle m \rangle$ that $\langle /m \rangle$ will expand its services division, Hewlett-Packard will acquire EYP Mission Critical Facilities*” and (2) “*HP to $\langle m \rangle$ Acquire $\langle /m \rangle$ Data Center Consultants.*” This categorization allowed us to identify cases where multimodal features are distinctly useful based on proportion of correctly

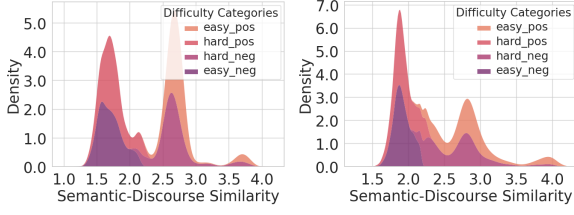


Figure 2: Kernel Density Estimation plots of semantic-discourse similarity scores (including Wu-Palmer similarity) for mention pair difficulty categories in ECB+ (L) and AIDA Phase 1 (R), showing a clear demarcation of easy and hard pairs in positive and negative labels. *easy_pos* and *hard_neg* pairs have a high semantic similarity distribution while *easy_neg* and *hard_pos* pairs have lower semantic similarity distribution.

resolved *hard* pairs (see Sec. 4).

Augmenting ECB+ with Images Since ECB+ does not provide images in their metadata, we scraped through the links provided in the documents and searched the Internet Archive for archived versions of articles with dead links. For original ECB documents without links, we manually search for keywords to retrieve articles. Out of 502 ECB+ document links, 43% were broken, but 50% could be recovered using `web.archive.org`. Of 480 ECB documents, 51% were located via Google search. We retrieved a total of 543 images; 235 of 982 documents had at least one associated image.

In addition to the overall lack of images, the retrieved document-level images may be poor representatives of individual event mentions, leading to the sparsity problem mentioned in Sec. 1. Therefore, we used Stable Diffusion (Rombach et al., 2022) to generate more relevant images and provide enough data to explore the contribution of multimodal information to ECR. Photo-realistic images were generated using sentences from ECB+ as prompts. Since a sentence can refer to multiple events, we provided an additional signal in the prompt by marking the event trigger with special tokens (`<m>` and `</m>`).

Image Encoding To encode all images as vector representations, we used three variations of Vision Transformers (ViT; Dosovitskiy et al. (2021), BEiT; Bao et al. (2021), and SWIN; Liu et al. (2021)), as well as CLIP (Radford et al., 2021). Resulting representations were the pooled output of the first-token representations from the last encoder layer for the image sequence, akin to the `[CLS]` token in BERT variants. Encoding the images through distinct embedding spaces decoupled them from the original language inputs.

Linear Projection Technique To project image and text representations across modalities, we first created a concatenated 3,072D (768×4) representation for an image/text pair. These concatenated representations contained the paired representation, the individual mention representations (`Arg1` and `Arg2`), and their element-wise product (in that order). Separate concatenated representations were constructed for each modality (see Fig. 1).² We then used a ridge regressor to calculate the linear coefficients by minimizing the squared distances between concatenated representations from each modality for the training set. This gave us two square ($3,072 \times 3,072$) “bridge” matrices: $M_{\mathcal{LLM} \rightarrow \mathcal{V}}$ and $M_{\mathcal{V} \rightarrow \mathcal{LLM}}$. We hypothesized that this bidirectional map retains crucial semantic information that a structure-preserving linear map would transfer between the two modalities. At evaluation, we matrix-multiplied the test concatenated representations with these matrices while maintaining the directionality of the linear map. These mapped representations were fed into a pairwise-scorer to get coreference clusters (see Fig. 1).

Model Training and Fine-Tuning Following Humeau et al. (2020); Cattani et al. (2021), i.e., we trained separate pairwise scorers $P_{\theta, \theta'}: (AB, BA) \rightarrow S_1, S_2$ on ECB+ and AIDA Phase 1. Here AB and BA are the 3,072D combined representations in $A \rightarrow B$ and $B \rightarrow A$ directions respectively, and θ and θ' are the parameters of the pairwise scorer and the LLM, respectively. This output two scores for each directional encoding, each representing the probability that the event mention pair was coreferent.³ Thereafter, we used the CoVal Scorer (Moosavi et al., 2019) to form the final coreference clusters after applying transitive closure to identify the connected components with a threshold of 0.5 for all models. We used the same pairwise-scorer for all linear maps.

For a direct multimodal comparison, we finetuned fused-modality models. We concatenated the image representations with the text representations and trained four separate pairwise scorers for each combination. Due to data sparsity of real images, we only trained fused models using generated event-centric images. Training took roughly 1.0 and 1.5 hours per epoch for the LLM and the fused models, respectively. For comparison, linear mapping took $\sim 3s$ to learn a mapping between modalities. Fig. 3 shows log GPU seconds re-

²All language representations came from the pre-trained Longformer model (Beltagy et al., 2020).

³As a human reader would likely make a consistent coreference decision regardless of which event description she read first, we used the mean of the two scores as the final probability score for training and inference.

quired for pairwise encoding for text, image, and fused modalities vs. bidirectional linear projection.

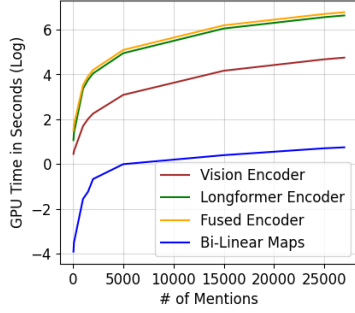


Figure 3: Pairwise encoding time in GPU seconds (log-scale on y-axis) for text (Longformer), vision (ViT), and fused models vs. Bidirectional Linear Mapping (Lin-Sem) as a function of the number of train pairs in ECB+.

4. Results and Analysis

We evaluate using established coreference metrics (Moosavi et al., 2019), e.g., MUC (Vilain et al., 1995), B^3 (Bagga and Baldwin, 1998), $CEAF_e$, and CoNLL F1 (the average of MUC, B^3 and $CEAF_e$ F1) scores.

4.1. ECB+

We present results from Held et al. (2021) as a current, commonly accepted SOTA on ECB+, and from Ahmed et al. (2023), whose computationally-efficient pruning heuristic based on surface lemma similarity we follow to allow us to perform multiple experiments on a smaller compute budget. Direct comparison to text-only model (LLM) performance should be taken as a comparison to Ahmed et al. (2023) due to the preprocessing. Table 2 shows detailed results.

Text-only vs. Multimodal Models Despite the extra training time incurred in training a fused-modality model with concatenated features (see Fig. 3), we see that the performance of the fused multimodal models does not exceed that of the text-only model (Longformer using Ahmed et al. (2023)’s preprocessing heuristic). Interestingly, the performance gap between linearly-mapped systems and fused modality models is often quite small, despite the higher compute cost of training the fused model. For instance, $\text{LLM} \rightarrow \text{BEiT-gen}$ and $\text{LLM} \rightarrow \text{BEiT-real}$ (Longformer embeddings mapped into BEiT space) slightly best the CoNLL F1 score of $\text{BEiT-gen} \oplus \text{LLM}$, and $\text{BEiT-real} \rightarrow \text{LLM}$ is only 0.5 F1 points lower. Similar trends hold when comparing other fused modality models and their linearly-mapped counterparts, such as $\text{LLM} \rightarrow \text{SWIN-gen}$, $\text{LLM} \rightarrow \text{SWIN-real}$, and $\text{SWIN-real} \rightarrow \text{LLM}$ vs. $\text{SWIN-gen} \oplus \text{LLM}$.

Models	MUC	B^3	$CEAF_e$	CoNLL
Held et al. (2021)	87.5	86.6	82.9	85.7
Ahmed et al. (2023)	90.8	86.7	84.7	87.4
ViT-real $\rightarrow$ LLM	6.9	63.1	55.1	41.7
BEiT-real $\rightarrow$ LLM	87.3	80.3	76.7	81.4
SWIN-real $\rightarrow$ LLM	87.6	79.7	76.5	81.3
CLIP-real $\rightarrow$ LLM	24.7	66.3	57.5	49.5
LLM $\rightarrow$ ViT-real	88.2	80.1	77.5	81.9
LLM $\rightarrow$ BEiT-real	88.3	80.0	77.4	81.9
LLM $\rightarrow$ SWIN-real	87.9	80.3	77.8	82.0
LLM $\rightarrow$ CLIP-real	88.3	80.0	77.4	81.9
ViT-gen $\oplus$ LLM	85.1	86.1	80.7	84.0
BEiT-gen $\oplus$ LLM	82.2	84.9	78.1	81.7
SWIN-gen $\oplus$ LLM	82.5	85.1	78.7	82.1
CLIP-gen $\oplus$ LLM	89.3	84.2	82.6	85.4
ViT-gen $\rightarrow$ LLM	77.4	78.8	71.5	75.9
BEiT-gen $\rightarrow$ LLM	77.8	79.8	73.7	77.1
SWIN-gen $\rightarrow$ LLM	79.5	79.6	73.4	77.5
CLIP-gen $\rightarrow$ LLM	83.0	82.1	76.3	80.5
LLM $\rightarrow$ ViT-gen	88.1	80.0	77.2	81.8
LLM $\rightarrow$ BEiT-gen	88.3	80.0	77.4	81.9
LLM $\rightarrow$ SWIN-gen	88.2	80.1	77.4	81.9
LLM $\rightarrow$ CLIP-gen	88.3	80.0	77.4	81.9

Table 2: MM-CDCR F1 scores for MUC, B^3 , $CEAF_e$ and CoNLL on ECB+ test set, using LLM only, Lin-Sem (“ $\rightarrow$ ”), and domain-fused finetuned versions (“ $\oplus$ ”). Cited works are previous benchmarks on text-only CDCR. **Bold** indicates the best performer on each metric. “-real” indicates that the vision space was encoded with real images, while “-gen” indicates generated images.

Semantic Transfer Categories F1 score obscures cases where different models may be performing better, and we see that although the aforementioned examples achieve comparable F1 scores, the linear mappings achieve a much higher MUC and B^3 recall, but lower precision, than the comparable fused models.⁴ Therefore, we do a proportional analysis of the correctly inferred (true positive) and misclassified (false positive and false negative) samples within the semantic transfer categories (see Table 3). These categorization labels were not used as supervision at any stage of training, fine-tuning, or mapping, and so an analysis of which models do better at which categories can illuminate different properties of the models, despite similar numerical performance. Table 3 shows the proportion of each result category per model, of samples that would be considered “hard” according to the mention pair difficulty categorization described in Sec. 3.

Within true positives (TP), linearly-mapped models, using both real and generated images, tended

⁴ $\text{LLM} \rightarrow \text{ViT-gen}$: MUC recall 98.7, precision 79.6; B^3 recall 97.6, precision 67.7. $\text{LLM} \rightarrow \text{BEiT-gen}$: MUC recall 99.1, precision 79.6; B^3 recall 97.9, precision 67.7. $\text{ViT-gen} \oplus \text{LLM}$: MUC recall 80.9, precision 89.7; B^3 recall 85.4, precision 86.9. $\text{BEiT-gen} \oplus \text{LLM}$: MUC recall 75.9, precision 89.7; B^3 recall 82.5, precision 87.5.

to correctly retrieve a higher proportion of hard pairs compared to the text-only and fused models. For instance, for generated images, the hard sample proportion retrieved by text-to-image models is almost 4 percentage points higher than that of text-only or fused models, while image-to-text models, though lower on average, still also correctly retrieve a higher proportion of hard pairs. This effect appears slightly more pronounced on average in the case of real images (avg. 51.8% hard pairs in TPs, compared to 50.1% for generated images, and 46.6% for text-only).

Semantic Transfer Categories				
Models	TP-Hard	FP-Hard	FN-Hard	
ECB+				
Ahmed et al. (2023)	0.466	0.521	0.607	
ViT-real→LLM	0.625	0.250	0.506	
BEiT-real→LLM	0.521	0.436	0.434	
SWIN-real→LLM	0.510	0.451	0.407	
CLIP-real→LLM	0.476	0.536	0.508	
LLM→ViT-real	0.507	0.456	0.441	
LLM→BEiT-real	0.506	0.000	0.000	
LLM→SWIN-real	0.496	0.438	0.700	
LLM→CLIP-real	0.505	0.452	0.708	
ViT-gen ⊕ LLM	0.432	0.591	0.635	
BEiT-gen ⊕ LLM	0.437	0.606	0.584	
SWIN-gen ⊕ LLM	0.404	0.620	0.642	
CLIP-gen ⊕ LLM	0.477	0.506	0.729	
ViT-gen→LLM	0.487	0.472	0.521	
BEiT-gen→LLM	0.471	0.445	0.525	
SWIN-gen→LLM	0.548	0.433	0.478	
CLIP-gen→LLM	0.483	0.490	0.534	
LLM→ViT-gen	0.505	0.449	0.541	
LLM→BEiT-gen	0.506	0.451	0.000	
LLM→SWIN-gen	0.505	0.452	0.531	
LLM→CLIP-gen	0.506	0.451	0.632	
AIDA Phase 1				
LLM	0.561	0.385	0.695	
ViT-real→LLM	0.609	0.368	0.734	
BEiT-real→LLM	0.661	0.328	0.629	
SWIN-real→LLM	0.660	0.327	0.636	
CLIP-real→LLM	0.627	0.332	0.657	
LLM→ViT-real	0.643	0.346	0.929	
LLM→BEiT-real	0.638	0.352	0.749	
LLM→SWIN-real	0.667	0.333	0.562	
LLM→CLIP-real	0.648	0.341	0.000	

Table 3: Table showing the proportion of hard event pairs within the true positive (TP), false positive (FP) and false negative (FN) samples based on semantic transfer category (Sec. 3) for ECB+. Values of 0 indicate that no cases fit this category, resulting in zero numerator.

Ensembling Models The apparent facility of different models at correctly retrieving mention pairs of different semantic difficulties led to a question: since the mention pair difficulty was never used during training, fine-tuning, or mapping, and only as an analytic tool, could we split the mention pairs according to their difficulty, and use the different model types to handle mention pairs they

on average appear to be better at? We therefore built an ensembling approach using the text-only model to handle easier pairs, and performed a grid-search through different combinations of the previously-trained multimodal models to handle harder pairs. We allowed for different multimodal models to potentially handle hard-positive pairs and hard-negative pairs and used the combined results from all models to compute the coreference metrics. Table 4 shows the best performing ensembles.

Models	MUC	B^3	$CEAF_e$	CoNLL
Held et al. (2021)	87.5	86.6	82.9	85.7
Ahmed et al. (2023)	90.8	86.7	84.7	87.4
ViT-gen ⊕ LLM + LLM	89.1	86.5	84.8	86.8
BEiT-gen ⊕ LLM + LLM	87.5	85.7	83.9	85.7
SWIN-gen ⊕ LLM + LLM	87.5	85.9	83.8	85.7
CLIP-gen ⊕ LLM + LLM	90.1	85.3	83.8	86.4
ViT-gen→LLM + LLM→BEiT-gen + LLM	90.8	85.2	84.8	86.9
BEiT-gen→LLM + LLM→BEiT-gen + LLM	91.3	85.5	86.5	87.8
SWIN-gen→LLM + LLM→BEiT-gen + LLM	90.4	84.4	83.8	86.2
CLIP-gen→LLM + LLM→BEiT-gen + LLM	91.2	85.3	85.7	87.4
LLM→ViT-gen + LLM→BEiT-gen + LLM	88.7	82.3	79.4	83.5
LLM→BEiT-gen + LLM	88.7	82.2	79.1	83.3
LLM→SWIN-gen + LLM→BEiT-gen + LLM	88.7	82.2	79.1	83.3
LLM→CLIP-gen + LLM→BEiT-gen + LLM	88.7	82.2	79.1	83.3
ViT-real→LLM + LLM→BEiT-real + LLM	94.5	89.5	91.8	91.9
BEiT-real→LLM + LLM→BEiT-real + LLM	88.9	82.4	79.7	83.7
SWIN-real→LLM + LLM→BEiT-real + LLM	88.7	82.2	79.1	83.3
CLIP-real→LLM + LLM→BEiT-real + LLM	94.3	89.3	91.6	91.7
LLM→ViT-real + LLM→BEiT-real + LLM	88.7	82.3	79.3	83.4
LLM→BEiT-real + LLM	88.7	82.2	79.1	83.3
LLM→SWIN-real + LLM→BEiT-real + LLM	89.0	82.7	80.1	83.9
LLM→CLIP-real + LLM→BEiT-real + LLM	88.7	82.2	79.1	83.3

Table 4: MM-CDCR MUC, B^3 , $CEAF_e$ and CoNLL F1 results on ECB+ test set, using ensemble models. Format follows Table 2. Ensemble model names follow the format *Hard-N model + Hard-P model + Easy pairs model*. LLM was always used to handle Easy pairs. The best performing models for hard negative and hard positives were found using a grid search through different combinations of multimodal models. If only one model besides LLM is listed, that model was used to handle all Hard pairs.

Our best performing ensemble model used ViT-real→LLM to handle hard negative pairs, LLM→BEiT-real, to handle hard positive pairs, and the text-only language model to handle easy pairs. This resulted in a CoNLL F1 score of 91.9, with scores of 89.5 or higher across all components of MUC, B^3 , or $CEAF_e$ metrics, showing the ability of this ensemble to score highly on, and balance, multiple measurements. Other ensembles, such as a variant that used CLIP-real→LLM to handle hard negatives, performed at a similar level. Two particularly interesting points emerge: 1) Using both real and generated images, LLM→BEiT routinely performed best at handling hard positive pairs; 2) Many ensemble models using Lin-Sem, especially those using a $\mathcal{V} \rightarrow \mathcal{LLM}$ mapping for hard negatives and an $\mathcal{LLM} \rightarrow \mathcal{V}$ mapping for hard positives, outperform the fused model/text-only model ensembles, despite the simplicity of the linear transformation. This suggests

that not only can visual information be leveraged for correct coreference of semantically more difficult mention pairs, but also that visual information may contain fine-grained cues useful for splitting mention pairs while linguistic information is more useful to cluster them.

4.2. AIDA Phase 1

Table 5 presents a novel baseline on the multi-modal AIDA Phase 1 data. This data contains unique challenges, such as a train set that is smaller than the test data, and event descriptions from sources with conflicting perspectives, explicitly addressing the ambiguity and perspective conflict challenges from Sec. 1. Since this data comes with images mappable to individual event mentions, we evaluate using only the provided images.

As with ECB+, we find that models using linear mappings compete with or slightly outperform the text only model. Using the same proportional analysis of correct and misclassified samples by difficulty category, we find that linearly-mapped models are also more likely than the text-only to resolve hard pairs correctly on this dataset (avg. hard pairs in TPs: 63.9% for $\mathcal{V} \rightarrow \mathcal{LLM}$, 64.9% for $\mathcal{LLM} \rightarrow \mathcal{V}$, and 56.1% for text-only).

We then applied the same ensembling approach to the AIDA data, using the same combination of linear mappings and the LLM according to the difficulty of the mention pair. Again we find that an ensemble model using a $\mathcal{V} \rightarrow \mathcal{LLM}$ mapping for hard negatives and an $\mathcal{LLM} \rightarrow \mathcal{V}$ mapping for hard positives performs best, although this time the model using CLIP-real $\rightarrow$ LLM as the hard negative handler comes out on top.

Models	MUC	B^3	$CEAF_e$	CoNLL
LLM	80.7	49.5	54.1	61.4
Vit-real $\rightarrow$ LLM	85.9	38.4	52.7	59.0
BEit-real $\rightarrow$ LLM	85.7	42.6	57.9	62.1
SWIN-real $\rightarrow$ LLM	82.9	46.4	55.8	61.7
CLIP-real $\rightarrow$ LLM	78.5	52.4	53.5	61.5
LLM $\rightarrow$ Vit-real	86.3	37.3	52.7	58.8
LLM $\rightarrow$ BEit-real	85.7	40.2	53.1	59.7
LLM $\rightarrow$ SWIN-real	86.2	39.1	54.4	59.9
LLM $\rightarrow$ CLIP-real	86.2	37.1	52.3	58.5
Vit-real $\rightarrow$ LLM + LLM $\rightarrow$ BEit-real + LLM	86.2	39.6	54.4	60.1
BEit-real $\rightarrow$ LLM + LLM $\rightarrow$ BEit-real + LLM	87.1	42.1	60.4	63.2
SWIN-real $\rightarrow$ LLM + LLM $\rightarrow$ BEit-real + LLM	87.1	42.5	60.5	63.4
CLIP-real $\rightarrow$ LLM + LLM $\rightarrow$ BEit-real + LLM	87.1	43.8	62.8	64.6
LLM $\rightarrow$ Vit-real + LLM $\rightarrow$ BEit-real + LLM	86.2	39.0	53.5	59.6
LLM $\rightarrow$ BEit-real + LLM	85.8	40.8	54.1	60.2
LLM $\rightarrow$ SWIN-real + LLM $\rightarrow$ BEit-real + LLM	86.6	40.7	56.6	61.3
LLM $\rightarrow$ CLIP-real + LLM $\rightarrow$ BEit-real + LLM	86.2	39.0	53.5	59.6

Table 5: MM-CDCR MUC, B^3 , $CEAF_e$ and CoNLL F1 results on AIDA Phase 1 Eval set. Format follows Tables 2 & 4. LLM denotes Longformer evaluated with Ahmed et al. (2023)’s methodology.

5. Discussion

Some specific example pairs where the text-only and fused models fail to link the pair, but ensembles correctly do so, expose certain features crucial for event coreference that are present in visual

information and linearly transferable, but missing in text alone or scrambled during model fusion.

ECB+ ECB+ examples of this kind include event pairs that require some sense of visual grounding, temporal logic (Schank and Abelson, 1975; Ravi et al., 2023) or pronominal context to resolve. For instance, pairs with pronominal antecedents and misleading lexical overlap like “...dozens of others were seriously injured in the quakes, which also sent small tsunamis...” and “...injured in the earthquakes which rekindled bitter memories of similar deadly quakes...”⁵ were missed by the LLM and fused models. Visual cues, such as damaged buildings or injured people (either in images generated using mentions as prompts, or already present in images in news articles) can help make the link. The aforementioned example is shown in Fig. 4, and the images are generated according to the ECB+ augmentation methodology (Sec. 3). Also in Fig. 4, Steven Moffat and his appear to be ambiguously overlapping to the text-only model, which missed the event mentions that are actually about Peter Capaldi. The two facial images, which are real images associated with the event mentions, help make the link.

AIDA Phase 1 Coreferent event mentions in the AIDA dataset are notable for conflicting information, and we find cases such as “Calling people tell about people that are jumping out of the burning building.” vs. “Forty-two people trapped by a fire on the third floor of the stately, Soviet-era Trades Unions building burned, suffocated or jumped to their deaths.” Text-only fails to link ambiguous event triggers, but the images associated with each show the Trades Unions building in Odesa. In such context-sensitive pairs, the paired visual representations (image domain Arg1 and Arg2 in Fig. 1) in Lin-Sem help resolve the coreference by capturing less ambiguous information from images while the text-only pairwise scorer found low contextual similarity between the event triggers. Similarly, we see pairs with ambiguous context or pronominal anaphora, e.g., “Buzina, 45, was shot dead” vs. “He was murdered”, are frequently missed by the LLM, but not by the ensemble models. In the case of this mention pair, both associated articles contain (different) pictures of the same individual, Oles Buzina, which, as with the ECB+ Peter Capaldi example, aids in the coreference⁶. Generally, for challenging corpora

⁵ “[E]arthquakes” vs. “quakes” is misleading lexical overlap as they refer to different earthquakes. The actual event triggers are underlined.

⁶ McNeely-White et al. (2022) present strong evidence for the particular effectiveness of linear transfor-



Figure 4: Sample coreferent event pairs from ECB+ that were correctly linked by our best multimodal ensemble ($ViT\text{-}real \rightarrow LLM + LLM \rightarrow BEiT\text{-}real + LLM$), but not by the text-only model. Event-triggers are highlighted in yellow and text in italics illustrates lexical ambiguity or misleading lexical overlap.

like AIDA Phase 1, we find visual features like faces, or background cues like angry protesters, press conferences, etc., act as cues for correctly resolving that pair.

6. Conclusion

In this paper we have demonstrated the utility of multimodal information in cross-document event coreference. In particular, our results demonstrate that multimodal information is useful for resolving mention pairs whose triggers have low semantic and discourse-level similarity, rendering them difficult for text-only models. We developed a method ($Lin\text{-}Sem$) for using linear transformations between embedding spaces to transfer semantic information between vision and language representation spaces, and used this technique in a model ensembling approach that used $Lin\text{-}Sem$ models to handle harder mention pairs and a text-only model for easier pairs. We applied this approach to the popular ECB+ benchmark and established a novel baseline on the challenging, and explicitly multimodal, AIDA Phase 1 dataset (Tracey et al., 2022). Our best performing models beat text-only performance on these datasets by ~ 3 F1

mation in face recognition.

points and establish an upper bound on CDCR performance given the preprocessing used. Our ablation studies show that ensemble systems built upon our mention pair difficulty categories and using structure preserving linear maps can leverage event-specific visual cues to make correct coreference decisions about difficult mention pairs. These visual cues are of course absent in text only models, and are likely scrambled during standard multimodal fusion approaches. As such, our results present a strong case for the utility of multimodal information in NLU tasks like event coreference and argue for future increased development of such resources. Upon publication, we will release our processing pipeline and the generated/scraped images associated with ECB+. ⁷

Our results should be considered in the context of our preprocessing assumptions. We use a computationally-efficient pruning heuristic that allowed us to run the high volume of experiments we showcased on a lower compute budget, while demonstrating the utility of multimodal features for coreference, especially for harder mention pairs. Our binary semantic transfer categories (*easy/hard*) do not account for semantic similarity between pairs that cross subtopics, or for singletons. As such, our current framework might not directly apply to corpora like FCC (Bugert et al., 2020), with cross-subtopic events.

7. Future Work

Future directions in this line of research include exploring the feasibility of using multimodal cues to align/enhance representation spaces of monolingual LLMs, like the English-only Longformer, for Russian and Ukrainian mention pairs in the AIDA Phase 1 corpus. Given the efficiency of linear transformations and the rarity of coreference-specific parallel corpora, this may help alleviate the compute budgets needed for multilingual LLM pretraining for CDCR. Another interesting direction is evaluating our method for other challenging CDCR datasets like FCC (Bugert et al., 2020) which contains cross-subtopic events or the GVC (Vossen et al., 2018) where the SOTA is lower compared to benchmarks like ECB+. Lastly, this work represents a novel cross-modal case where affine transformations between embedding spaces has been shown to be useful (cf. McNeely-White et al. (2022); Nath et al. (2022); Merullo et al. (2023); Ghaffari and Krishnaswamy (2023)). Future work in this area entails a theoretical exploration of the properties of embedding spaces with a goal of finding performance guarantees where affine transformations successfully preserve information for different AI tasks.

⁷The AIDA Phase 1 data must be properly obtained from the Linguistic Data Consortium.

Ethics Statement

Our ablation studies required a non-trivial computation budget and concomitant resource usage, especially for the fused models with larger scoring heads on top of the LLM. Moreover, even though our `Lin-Sem` framework is substantially compute-efficient, it still required cross-modal model encoding in generating representations for deploying our linear maps between them. The images generated for this task with diffusion models might reflect social, racial, or gender-based stereotypes as are commonly seen in large generative models. Due to the nature of the AIDA Phase 1 data’s focus on Ukrainian-Russian conflict, the events described therein are likely to be distressing to some.

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